tf.compat.v1.math.reduce_sum Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/math/reduce_sum

> *Computes the sum of elements across dimensions of a tensor. (deprecated arguments)*

Function Signatures

```
tf.compat.v1.math.reduce_sum( input_tensor, axis=None,
keepdims=None, name=None, reduction_indices=None,
keep_dims=None )
(keep_dims)
```

Code Examples

```
tf.compat.v1.math.reduce_sum(
input_tensor,
axis=None,
keepdims=None,
name=None,
reduction_indices=None,
keep_dims=None
)
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input_tensor,
axis=None,
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)
```

```
>>> # x has a shape of (2, 3) (two rows and three columns):
>>> x = tf.constant([[1, 1, 1], [1, 1, 1]])
>>> x.numpy()
array([[1, 1, 1],
[1, 1, 1]], dtype=int32)
>>> # sum all the elements
>>> # 1 + 1 + 1 + 1 + 1+ 1 = 6
>>> tf.reduce_sum(x).numpy()
6
>>> # reduce along the first dimension
>>> # the result is [1, 1, 1] + [1, 1, 1] = [2, 2, 2]
>>> tf.reduce_sum(x, 0).numpy()
array([2, 2, 2], dtype=int32)
>>> # reduce along the second dimension
>>> # the result is [1, 1] + [1, 1] + [1, 1] = [3, 3]
>>> tf.reduce_sum(x, 1).numpy()
array([3, 3], dtype=int32)
>>> # keep the original dimensions
>>> tf.reduce_sum(x, 1, keepdims=True).numpy()
array([[3],
[3]], dtype=int32)
>>> # reduce along both dimensions
>>> # the result is 1 + 1 + 1 + 1 + 1 + 1 = 6
>>> # or, equivalently, reduce along rows, then reduce the
resultant array
>>> # [1, 1, 1] + [1, 1, 1] = [2, 2, 2]
>>> # 2 + 2 + 2 = 6
>>> tf.reduce_sum(x, [0, 1]).numpy()
6
```

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array([[1, 1, 1],
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6
```

Documentation

Computes the sum of elements across dimensions of a tensor. (deprecated arguments)

This is the reduction operation for the elementwise tf.math.add op.

Reduces input_tensor along the dimensions given in axis.
Unless keepdims is true, the rank of the tensor is reduced by 1 for each
of the entries in axis, which must be unique. If keepdims is true, the
reduced dimensions are retained with length 1.

If axis is None, all dimensions are reduced, and a
tensor with a single element is returned.

For example

Returns

numpy compatibility

Equivalent to np.sum apart the fact that numpy upcast uint8 and int32 to
int64 while tensorflow returns the same dtype as the input.